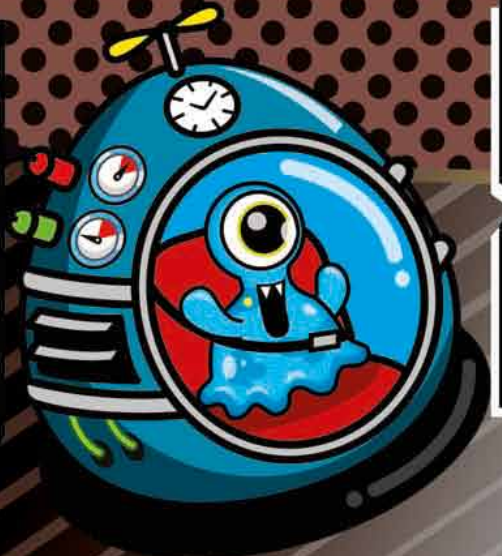
IS TIME TRAVEL A POSSIBILITY?

We all travel in time, at the rate of one hour per hour. Duh! But could we travel faster or slower than that? Could it be possible to travel backward and forward in time? Think what a difference that would make. If you have homework due tomorrow, how about going back in time two days so that you have more time to finish it, then zipping forward to the present and handing it in early? It's time to investigate.

SPEED LIMITS

Imagine that you were in a spaceship traveling near the speed of light, which is much faster than it is possible to go now. When you returned to Earth, all of your friends would be much older. That's not because you hang around old people! It's because time has passed more slowly for you than those you left behind. In a way, you have been a time traveler. Einstein figured out that if we could ever break the speed limit, we could travel through time. Sadly, we're slowpokes and we can't.



BLACK HOLES AND WORMHOLES

Einstein had another theory of general relativity that said that time passes more slowly for objects held in gravitational fields, such as Earth, than for objects far away from the fields. Black holes are areas in space where the gravity is so intense that it sucks in everything, even light and potential time travelers. Black holes could lead to wormholes, tunnels that act as shortcuts through space and time. Hole-y moley!

EINSTEIN'S THEORY

In his famous theory of relativity, big-thinking German physicist Albert Einstein (1879–1955) proposed that space and time are aspects of the same thing: space-time. There is a speed limit of precisely 186,282 miles (299,792 km) per second for anything that travels through space-time. Light always obeys the rules and travels through empty space at the speed limit. Einstein deduced that you need to travel faster than light to travel through space-time.

LAWS OF PHYSICS

In 1895, English author H. G. Wells (1866–1946) wrote a book called *The Time Machine* about a vehicle that could take people through time. It might sound far-fetched, but physicists have a saying—if nothing is prohibited, it must happen at some point. There is nothing in the laws of physics that prohibits time travel. The prospect of travel into the future seems possible, because time passes at a different rate in orbit than it does on the ground. Even though there are no laws to prohibit travel into the past, most physicists agree that it is probably less likely to happen.

"ONCE CONFINED TO FANTASY AND SCIENCE FICTION, TIME TRAVEL IS NOW SIMPLY AN ENGINEERING PROBLEM."
Michio Kaku (1947–),
American physicist